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import java.util.*;
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public class Multiplicative_Cypher {
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    static int gcd(int a, int b) {
        while (b != 0) {
            int temp = b;
            b = a % b;
            a = temp;
        }
        return a;
    }

    public static String Encryption() {

        Scanner sc = new Scanner(System.in);
        int key;
        char[] text;

        while (true) {
            System.out.print("Enter the Text in LowerCase : ");
            String str = sc.nextLine();

            text = str.toCharArray();
            boolean valid = true;

            for (int i = 0; i < text.length; i++) {

                if (text[i] == ' ') {
                    continue;
                }
                else if (text[i] < 'a' || text[i] > 'z') {
                    valid = false;
                    break;
                }
            }

            if (!valid) {
                System.out.println(
                    "Please Enter text only and also make sure it is in lower Case:"
                );
                continue;
            }
            else {
                break;
            }
        }

        while (true) {
            System.out.print("Enter the Key : ");
            key = sc.nextInt();
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        if (gcd(key, 26) == 1) {
            break;
        }

        System.out.println("Invalid Key");
    }

    char cypher[] = new char[text.length];

    for (int i = 0; i < text.length; i++) {

        if (text[i] == ' ') {
            cypher[i] = text[i];
            continue;
        }

        char a = text[i];

        int n = (int) a - 97;
        int t = ((n * key) % 26);

        cypher[i] = (char) (t + 97);
    }

    String cyphertext = new String(cypher);
    cyphertext = cyphertext.toUpperCase();

    return cyphertext;
}

public static String Decryption() {

    Scanner sc = new Scanner(System.in);
    int key;
    char[] text;

    System.out.print("Enter The Encrypted text In UpperCase: ");

    String str = sc.nextLine();
    str = str.toLowerCase();

    text = str.toCharArray();

    while (true) {

        System.out.print("Enter The Key Used In Encryption: ");
        key = sc.nextInt();

        if (gcd(key, 26) == 1) {
            break;
        }

        System.out.println("Invalid Key");
    }
}

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}

char cypher[] = new char[text.length];

for (int i = 0; i < text.length; i++) {

    if (text[i] == ' ') {
        cypher[i] = ' ';
        continue;
    }

    char a = text[i];

    int n = (int) a - 97;
    int na = 0;

    for (int j = 1; j <= 26; j = j + 2) {

        if (j == 13) {
            continue;
        }

        if ((key * j) % 26 == 1) {
            na = j;
            break;
        }
    }

    int t = ((n * na) % 26);

    cypher[i] = (char) (t + 97);
}

String cyphertext = new String(cypher);
cyphertext = cyphertext.toLowerCase();

return cyphertext;
}

public static int Brute_force(String x, String y) {

    char e[] = x.toCharArray();
    char c[] = new char[e.length];

    int key = -1;

    for (int k = 1; k < 26; k = k + 2) {

        if (k == 13) {
            continue;
        }

        for (int i = 0; i < e.length; i++) {

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        if (e[i] == ' ') {
            c[i] = ' ';
            continue;
        }

        char a = e[i];

        int n = (int) a - 97;
        int na = 0;

        for (int j = 1; j <= 26; j = j + 2) {

            if (j == 13) {
                continue;
            }

            if ((k * j) % 26 == 1) {
                na = j;
                break;
            }
        }

        int t = ((n * na) % 26);

        c[i] = (char) (t + 97);
    }

    String s = new String(c);

    if (y.equals(s)) {
        key = k;
        return key;
    }
}

System.out.print(key);

return -1;
}

public static void main(String s[]) {

    while (true) {

        Scanner sc = new Scanner(System.in);

        System.out.print(
            "For Encryption : 1 " +
            "\t For Decryption : 2" +
            "\nFor Bruteforce : 3" +
            "\t For Exit : 99" +
            "\nEnter Your Choice : "

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    );

    int a = sc.nextInt();

    if (a == 1) {

        System.out.println(
            "This is The Encrypted Text : " + Encryption()
        );
    }
    else if (a == 2) {

        System.out.println(
            "This is Decrypted Text : " + Decryption()
        );
    }
    else if (a == 3) {

        System.out.print("So First Give the Encrypted Key: ");

        sc.nextLine();

        String x = sc.nextLine();

        System.out.print("Enter The Decrypted Code :");

        String y = sc.nextLine();

        x = x.toLowerCase();

        int r = Brute_force(x, y);

        System.out.println("Key is: " + r);
    }
    else if (a == 99) {
        break;
    }
}
}

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}

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